STAT 400 Homework 5: Random Variables

1) An appliance store receives a shipment of 30 ovens, 5 of which are defective (unknown to the manager). The store manager selects 4 ovens at random, without replacement, and tests to see if they are defective. Let X be the number of defective ovens found in the sample of 4.

   Calculate the probability mass function (PMF) of X.

$$P_X(0) = P(X = 0) = \frac{\binom{5}{0} \cdot \binom{25}{4}}{\binom{30}{4}}$$

$$P_X(0) = \mathbf{0.462}$$

$$P_X(1) = P(X = 1) = \frac{\binom{5}{1} \cdot \binom{25}{3}}{\binom{30}{4}}$$

$$P_X(1) = \mathbf{0.420}$$

$$P_X(2) = P(X = 2) = \frac{\binom{5}{2} \cdot \binom{25}{2}}{\binom{30}{4}}$$

$$P_X(2) = \mathbf{0.109}$$

$$P_X(3) = P(X = 3) = \frac{\binom{5}{3} \cdot \binom{25}{1}}{\binom{30}{4}}$$

$$P_X(3) = \mathbf{0.00912}$$

$$P_X(4) = P(X = 4) = \frac{\binom{5}{4} \cdot \binom{25}{0}}{\binom{30}{4}}$$

$$P_X(4) = \mathbf{0.000182}$$

2) Consider the function

$$p(x) = \frac{e^{-5}5^x}{x!}, \quad x \in \{0, 1, 2, 3, 4, 5, \dots\}.$$

a) Show that $p(x)$ satisfies the conditions to be a probability mass function.

PMF Property 1

e^{-5} is a positive constant. Additionally, 5^x and $x!$ are positive for any value of x. Thus, $p(x) \geq 0$.

PMF Property 2

Let $\mathcal{X} = \{0, 1, 2, 3, 4, 5, \dots\}$.

$$\sum_{x \in \mathcal{X}} p(x) = \sum_{x=0}^{\infty} \frac{e^{-5}5^x}{x!}$$
$$\sum_{x \in \mathcal{X}} p(x) = e^{-5} \sum_{x=0}^{\infty} \frac{5^x}{x!}$$
$$\sum_{x \in \mathcal{X}} p(x) = e^{-5}e^{5}$$
$$\sum_{x \in \mathcal{X}} p(x) = 1$$

$\therefore$ $p(x)$ is a probability mass function.

b) Let

$$Y = 3^X.$$

Calculate the PMF of Y.

$$P_Y\left(3^x\right) = P\left(Y = 3^x\right) = P(X = x)$$
$$P_Y\left(3^x\right) = p(x)$$
$$P_Y\left(3^x\right) = \boldsymbol{\frac{e^{-5}5^x}{x!}}$$

c) Calculate the expected value of Y, that is, $E(Y)$.

Let $\mathcal{X} = \{0, 1, 2, 3, 4, 5, \dots\}$

$$E(Y) = \sum_{x \in \mathcal{X}} 3^x \frac{e^{-5}5^x}{x!} = \sum_{x=0}^{\infty} 3^x \frac{e^{-5}5^x}{x!}$$
$$E(Y) = e^{-5} \sum_{x=0}^{\infty} \frac{15^x}{x!}$$
$$E(Y) = e^{-5}e^{15} = \boldsymbol{e^{10}}$$

3) Suppose

$$f(x) = \begin{cases} \frac{1}{(x+1)^6}, & \text{if } x \in (0, \infty), \\ 0, & \text{else.} \end{cases}$$

a) Find the constant c so that $cf(x)$ satisfies the conditions to be a probability density function of a random variable, say X. Use Desmos to draw the graph of the density function $f_X(x)$.

PDF Property 1

$$\begin{aligned} cf(x) &\geq 0 \\ \frac{c}{(x+1)^6} &\geq 0 \\ \therefore c &\geq 0 \end{aligned}$$

PDF Property 2

$$\begin{aligned} \int_0^\infty cf(x)\,\mathrm{d}x &= 1 \\ c\int_0^\infty \frac{1}{(x+1)^6}\,\mathrm{d}x &= 1 \\ c\left(-\frac{1}{5(x+6)^5}\right)\Big|_0^\infty &= 1 \\ \lim_{b\to\infty}\left(c\left(-\frac{1}{5(x+1)^5}\right)\Big|_0^b\right) &= 1 \\ \lim_{b\to\infty}\left(c\left(-\frac{1}{5(b+1)^5} + \frac{1}{5(1)^5}\right)\right) &= 1 \\ \frac{c}{5} &= 1 \\ \boldsymbol{c} &= \mathbf{5} \end{aligned}$$

3) b) Calculate the expected value $E(X)$ and the CDF $F_X(x)$.

$$E(X) = 5\int_0^\infty x f_X(x)\,\mathrm{d}x$$

$$u = x + 1$$
$$\mathrm{d}u = \mathrm{d}x$$

$$E(X) = 5\int_1^\infty \frac{u-1}{u^6}\,\mathrm{d}u$$

$$E(X) = 5\int_1^\infty \left(\frac{1}{u^5} - \frac{1}{u^6}\right)\mathrm{d}u$$

$$E(X) = 5\left(-\frac{1}{4u^4} + \frac{1}{5u^5}\right)\Bigg|_1^\infty$$

$$E(X) = \lim_{b\to\infty}\left(5\left(-\frac{1}{4u^4} + \frac{1}{5u^5}\right)\Bigg|_1^b\right)$$

$$E(X) = \lim_{b\to\infty}\left(5\left(-\frac{1}{4b^4} + \frac{1}{5b^5} + \frac{1}{4} - \frac{1}{5}\right)\right)$$

$$E(X) = \frac{5}{4} - \frac{5}{5}$$

$$E(X) = \mathbf{0.25}$$

$$F_X(x) = \int_0^x 5f(t)\,\mathrm{d}t$$

$$F_X(x) = 5\int_0^x \frac{\mathrm{d}t}{(t+1)^6}$$

$$F_X(x) = 5\left(-\frac{1}{5(t+1)^5}\right)\Bigg|_0^x$$

$$F_X(x) = -\frac{5}{5(x+1)^5} + \frac{5}{5}$$

$$F_X(x) = \mathbf{1} - \frac{\mathbf{1}}{(\boldsymbol{x}+\mathbf{1})^{\mathbf{5}}}$$

3) c) Calculate the pth percentiles for

$$p = 0.2,\ 0.5,\ 0.8,\ 0.99.$$

$$F_X(n_p) = p$$

$$1 - \frac{1}{(n_p + 1)^5} = p$$

$$1 - p = \frac{1}{(n_p + 1)^5}$$

$$(n_p + 1)^5 = \frac{1}{1 - p}$$

$$n_p = \frac{1}{(1-p)^{1/5}} - 1$$

$$n_{0.2} = \frac{1}{(1-0.2)^{1/5}} - 1$$

$$n_{0.2} = \mathbf{0.0456}$$

$$n_{0.5} = \frac{1}{(1-0.5)^{1/5}} - 1$$

$$n_{0.5} = \mathbf{0.149}$$

$$n_{0.8} = \frac{1}{(1-0.2)^{1/5}} - 1$$

$$n_{0.8} = \mathbf{0.380}$$

$$n_{0.99} = \frac{1}{(1-0.2)^{1/5}} - 1$$

$$n_{0.99} = \mathbf{1.512}$$

d) Calculate the αth critical value for

$$\alpha = 0.05.$$

$$P(X \geq x_{0.05}) = 0.05$$

$$F_X(x_{0.05}) = 1 - 0.05$$

$$1 - \frac{1}{(x_{0.05} + 1)^5} = 0.95$$

$$0.05 = \frac{1}{(x_{0.05} + 1)^5}$$

$$(x_{0.05} + 1)^5 = 20$$

$$x_{0.05} = \mathbf{0.821}$$

3) e) The random variable X is a special case of the Pareto distribution, which is used to describe the "Pareto Principle," often used to model wealth, the impact of different factors on sales, and even online dating trends.

The principle broadly suggests that the majority of the impact, wealth, or ownership is often determined by a small (yet influential) set of actions or people. Calculate

$$P(X \leq 1),$$

and use this to give an intuitive explanation of the Pareto Principle.

$$P(X \leq 1) = F_X(1)$$

$$P(X \leq 1) = 1 - \frac{1}{(1+1)^5}$$

$$P(X \leq 1) = 0.96875$$

X assuming any value in $(0, 1]$ is already very likely because $P(X \leq 1) = 0.96875$ despite $x \in (0, \infty)$.

4) Suppose

$$f(x) = \begin{cases} x^2(1-x)^2, & \text{if } x \in (0,1), \\ 0, & \text{else.} \end{cases}$$

a) Find the constant c so that $cf(x)$ satisfies the conditions to be a probability density function of a random variable, say X.

PDF Property 1

$$\begin{aligned} cf(x) &\geq 0 \\ cx^2(1-x)^2 &\geq 0 \\ \therefore c &\geq 0 \end{aligned}$$

PDF Property 2

$$\begin{aligned} \int_0^1 cf(x)\,\mathrm{d}x + 0 &= 1 \\ c\int_0^1 x^2(1-x)^2\,\mathrm{d}x &= 1 \\ c\int_0^1 \left(x^4 - 2x^3 + x^2\right)\mathrm{d}x &= 1 \\ c\left(\frac{1}{5}x^5 - \frac{1}{2}x^4 + \frac{1}{3}x^3\right)\Big|_0^1 &= 1 \\ c\left(\frac{1}{5} - \frac{1}{2} + \frac{1}{3}\right) &= 1 \\ \frac{c}{30} &= 1 \\ \boldsymbol{c} &= \mathbf{30} \end{aligned}$$

4) b) Calculate $E(X)$.

$$E(X) = \int_0^1 30x f(x)\,\mathrm{d}x$$

$$E(X) = 30\int_0^1 x^3(1-x)^2\,\mathrm{d}x$$

$$E(X) = 30\int_0^1 \left(x^5 - 2x^4 + x^3\right)\mathrm{d}x$$

$$E(X) = 30\left(\frac{1}{6}x^6 - \frac{2}{5}x^5 + \frac{1}{4}x^4\right)\Bigg|_0^1$$

$$E(X) = 30\left(\frac{1}{6} - \frac{2}{5} + \frac{1}{4}\right)$$

$$E(X) = \mathbf{\frac{1}{2}}$$

c) The distribution of X can be used to model proportions. For example, we could assume that X models the proportion of students on campus who are computer science majors.

Under this assumption, calculate and interpret the exact value of

$$P(0.2 \leq X \leq 0.5).$$

$$P(0.2 \leq X \leq 0.5) = \int_{0.2}^{0.5} 30 f(t)\,\mathrm{d}t$$

$$P(0.2 \leq X \leq 0.5) = 30\int_{0.2}^{0.5} x^2(1-x)^2\,\mathrm{d}t$$

$$P(0.2 \leq X \leq 0.5) = 30\int_{0.2}^{0.5} \left(x^4 - 2x^3 + x^2\right)\mathrm{d}x$$

$$P(0.2 \leq X \leq 0.5) = 30\left(\frac{1}{5}x^5 - \frac{1}{2}x^4 + \frac{1}{3}x^3\right)\Bigg|_{0.2}^{0.5}$$

$$P(0.2 \leq X \leq 0.5) = 0.44208$$

The value 0.44208 is the probability that anywhere from 20% to 50% of the students on campus are computer science majors.

5) We toss a coin with $P(H) = p \in (0, 1)$ indefinitely and independently. You win 3^n dollars if, on a given run of the coin tosses, H shows up for the first time on the nth toss.

Let X be the random variable that gives your winnings for one run of the game.

a) Calculate the probability mass function of X.

$$P_X\left(X = 3^n\right) = \boldsymbol{p(1-p)^{n-1}}$$

b) If you play this game repeatedly, what would your expected winnings, $E(X)$, be? Note that, depending on the value of p, $E(X)$ can be finite or infinite.

$$E(X) = \sum_{n=1}^{\infty} 3^n p(1-p)^{n-1}$$

$$E(X) = \frac{p}{1-p} \sum_{n=1}^{\infty} 3^n (1-p)^n$$

$$E(X) = \frac{p}{1-p} \sum_{n=1}^{\infty} (3-3p)^n$$

$$E(X) = \frac{p}{1-p} \cdot \frac{3-3p}{1-(3-3p)}$$

$$E(X) = \boldsymbol{\frac{3p}{3p-2}} \textbf{ where } \boldsymbol{\frac{2}{3} < p < 1}$$

c) Suppose you could play this game indefinitely. What is the most you would be willing to pay to play this game so that your expected net earnings are at least 5 dollars? Your answer will depend on the value of p (when $E(X)$ is finite).

$$\text{Fee} = E(X) - 5$$

$$\text{Fee} = \frac{3p}{3p-2} - 5$$

$$\text{Fee} = \frac{3p}{3p-2} - \frac{15p-10}{3p-2}$$

$$\text{Fee} = \boldsymbol{\frac{10-12p}{3p-2}} \textbf{ where } \boldsymbol{\frac{2}{3} < p < \frac{5}{6}}$$